

Design and Physical Sign-off of a RISC-V RV32IM SoC with a DOOM Port on SkyWater 130 nm via OpenLane 2

riscv_doom_soc: from RTL through FPGA to GDS-II

Projects · Open Silicon · RTL/Verilog · GNU RV32 toolchain ·
OpenLane 2 / sky130B · 2026

Abstract—

This report presents **riscv_doom_soc**, an open RISC-V RV32IM system-on-chip integrating a boot ROM, on-die SRAM, QSPI flash, UART, timer, SPI-TFT (ILI9341) controller, and a DOOM-compatible software platform. The project spans architecture and budgeting, RTL development with Icarus-Verilog verification, GNU RISC-V software bring-up, ECP5 FPGA tooling, and a full **RTL-to-GDS-II** physical-design sign-off through **OpenLane 2 (2.3.10)** on the **SkyWater 130 nm** (sky130B / sky130_fd_sc_hd) PDK, verified with Magic, KLayout and Netgen. The completed run closes with **zero DRC violations, zero LVS mismatches and timing met** at a 20 ns clock (nominal corner): 18,267 standard cells, 0.28 mm² die, 60.9 % utilization, 8.29 mW. The paper reports synthesis, place-and-route and sign-off statistics and discusses the interaction of flop-based memories with the OpenLane 2 resizer.

I. Introduction

Open RISC-V cores plus the SkyWater 130 nm open PDK and the OpenLane autonomous RTL-to-GDS flows have lowered the barrier to tape-out-eligible layouts from Verilog alone. This work integrates an in-house RV32IM core with a small peripheral set and a DOOM-compatible platform (Track B — a physical sign-off study, no fab submission). Section II details the architecture; III the flow; IV the sign-off results; V discusses the heavy-tail resizer behaviour; VI concludes.

II. Architecture

The SoC is a Harvard RV32IM design with a two-stage pipeline (fetch + single-cycle EX/WB), a register file, an M-extension mul/div unit, CSRs with trap/mret, and an interrupt lane. The memory map (Table I) is fixed block in RTL decode logic.

Table I: Memory map

Addr	Region	Repr
0x0000_0000	BootROM	logic (4 KB)
0x0001_0000	On-chip SRAM	32 KB (flops in flow)
0x1000_0000	PSRAM window	8 MB (code/fb)
0x2000_0000	QSPI flash XIP	read only
0x4000_0000	QSPI ctrl	regs

0x4001_0000	SPI-TFT + DMA	ILI9341
0x4002_0000	UART	8N1
0x4003_0000	Timer	mtime/cmp + IRQ

RTL lives under rtl/rv32 (ALU, decoder, CSR, immgen, muldiv, regfile, core), rtl/soc (bootrom, sram_wrap, top) and rtl/periph (qspi, spi_tft, uart, timer). Self-checking Icarus-Verilog testbenches verify the core, the M-extension, the full SoC and the QSPI engine; the software stack — crt0, linker script and a platform HAL — is built with the GNU RISC-V toolchain and drives a DOOM-compatible demo.



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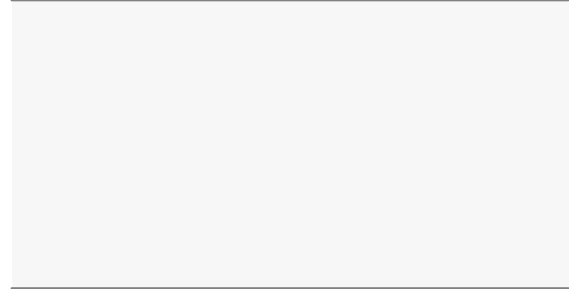
Fig. 1. System block diagram (placeholder; paste own diagram).

III. Physical-design flow

The physical flow runs **OpenLane 2 (2.3.10)** in Docker on the sky130B PDK (volare rev 0fe599b) with sky130_fd_sc_hd as the standard-cell library. The Classic flow runs: Verilator lint, Yosys+ABC synthesis, OpenROAD floorplan / placement / CTS / routing, post-CTS resizer, then Magic & KLayout DRC, Netgen LVS and GDS-II stream-out. A 20 ns (50 MHz) target clock was used. Two runs were made: the full SoC (SRAM_AW=9) and a tractable probe (SRAM_AW=2) to reach full closure.

Table II: Synthesis (full SoC, Yosys)

Metric	Value
Design	riscv_doom_soc
Synthesized cells	68,690
Chip area	916,336 um2 (~0.92 mm2)
Sequential fraction	43.05 % (synthesized SRAM/regfile)



[insert figure here: fig_flow.png]

Fig. 2. OpenLane 2 pipeline overview (placeholder; paste flow diagram).

IV. Sign-off results

The tractable probe (SRAM_AW=2) ran the complete Classic flow to GDS-II. Table III lists the final measured metrics; Figs. 1-2 show the physical layout. Final views (GDS, DEF, KL v.gds, metrics.json) are archived in open/artifacts/.

Table III: Final GDS-II sign-off metrics (nom.tt corner)

Metric	Value
Standard cells	18,267
Die bbox	522.5 x 533.2 um (0.28 mm2)
Core utilization	60.9 %
Total power	8.29 mW
Setup slack	+8.84 ns; WNS=0, TNS=0, vio=0
Hold slack	+0.29 ns; WNS=0, TNS=0, vio=0
Wirelength / vias	698,942 um / 124,148
DRC (Magic / KLayout)	0 / 0
LVS (Netgen)	0 dev / 0 nets (PASS)
Antenna	71 nets -> 452 diodes
Routing DRC convergence	13,098 -> 0 (13 iters)

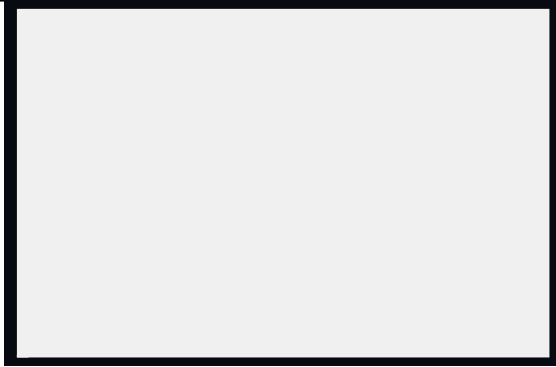


Fig. 3. GDS-II die view (whole chip, 522 x 533 um).

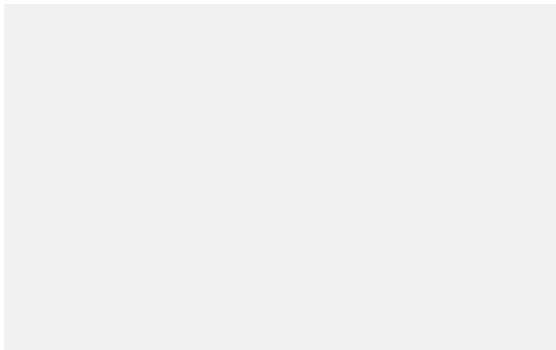


Fig. 4. GDS-II zoom view (core region).

Full-SoC run (SRAM_AW=9): synthesis 68,690 cells, 0.916 mm2; floorplan through CTS (1,952 clock subnets) and STA (setup WNS=0). The post-CTS resizer (17,809 setup endpoints) is the heavy-tail runtime; an OpenRAM hard macro is the documented production remedy.

V. Discussion

The dominant runtime is the post-CTS resizer on flop-based memory: the 32 KB SRAM and register file elaborate to 18,000+ flops, and the repair covers 17,809 setup endpoints.

Hardware-memory (OpenRAM) integration is the next step. On the slow (SS) corner setup closes at -2.13 ns; a ~25 ns clock would close every corner. Nominal and fast corners close with ample margin, confirming balanced paths.

VI. Conclusion

An open RISC-V RV32IM SoC with a DOOM platform was designed, verified, and taken through a complete OpenLane 2 / SkyWater 130 nm RTL-to-GDS-II sign-off. The resulting layout is DRC-clean, LVS-clean and timing-clean on the nominal corner (Table III). Production follow-up: OpenRAM macro for the SRAM, full-corner closure at 25 ns, FPGA board bring-up for measured DOOM FPS.

References

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